European Polymer Journal 162 (2022) 110897

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

HCN-derived polymers from thermally induced polymerization of diaminomaleonitrile: A non-enzymatic peroxide sensor based on prebiotic chemistry

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ARTICLE INFO

Keywords:
HCN-derived polymers
DAMN polymerization
Electrocatalysis
Biosensors
Multifunctional materials
Prebiotic chemistry

ABSTRACT

HCN-derived polymers have recently attracted considerable attention due to their promising applications as multifunctional materials. This study, inspired by plausible early Earth geochemical conditions, describes a strategy to synthesize them from the self-initiated thermal bulk polymerization of the HCN tetramer, diaminomaleonitrile (DAMN), with outstanding sensing properties. These conjugated polymers were obtained through noncatalysed and simple isothermal reactions at 170 °C in the solid-state, and experiments at 190 °C permitted polymerization in the melt. Both processes are highly efficient, allowing quantitative yields of the end products. The conductivity properties of both polymers have been explored to show their high potential, especially DAMN polymers synthesized in melt, as nonenzymatic peroxide sensors. To better understand the differences found between the two series, structural characterisation was carried out using compositional data, Fourier transform infrared (FTIR), nuclear magnetic resonance (NMR) and X-ray photoelectron (XPS) spectroscopies, and X-ray diffraction (XRD) measurements. The interpretation of the structural data suggests that a two-dimensional (2-D) macrostructure based on N-heterocyclics is predominant regardless of the state of monomer aggregation during the course of polymerization, but preferably formed in the melt. The morphological and thermal stability properties of the polymers based on DAMN were also evaluated. Finally, the most likely mechanisms based on the dehydrocyanation and deamination reactions that take place during the polymerization reaction are proposed. This study demonstrates the robust and straightforward character of these thermally activated polymerizations, which are of interest to chemical evolution research and to current materials and surface science.

1. Introduction

HCN-derived polymers, or just HCN polymers, have received increasing attention in materials science due to their potential for the development of a new generation of multifunctional materials [1,2]. They comprise a heterogeneous family of complex polymeric substances synthetized from net HCN, its alkaline salts as KCN/NaCN, also NH_4CN, from its trimer aminomalononitrile (AMN) or its tetramer diaminomaleonitrile (DAMN), and from formamide, its hydrolysis product. All these precursors of HCN-derived polymers can be generated easily abiotically under numerous terrestrial or extraterrestrial environmental conditions; subsequently, some conditions allow their polymerization [3]. Thus, prebiotic chemistry has recently inspired novel polymeric HCN-based films with potential applications in biomedicine [4–10] or as protective coatings against oxidation [11–13].

Based on these previous results and considering that the thermally induced polymerization of DAMN might be possible under the extreme terrestrial and thalassic thermal/hydrothermal conditions of primordial Earth such as hydrothermal vents close to volcanoes [14], here DAMN polymers will be synthetized under comparable conditions. Polymerizations with commercial DAMN were isothermally initiated at 170 and 190 °C under anoxic conditions. These specific temperatures were chosen taking into account very recent kinetic studies and preliminary structural characterization results of DAMN polymerization in the solid-

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https://doi.org/10.1016/j.eurpolymj.2021.110897
Received 6 October 2021; Received in revised form 17 November 2021; Accepted 22 November 2021
Available online 24 November 2021
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state at 170 °C [15] and in the melt at 190 °C [16]. The final products from these bulk polymerizations then will be assayed as materials to modify electrodes. The good electrochemical properties observed, especially for the DAMN polymer synthesized at the higher temperature, have led to research on its potential as a basis for biosensors. Specifically, we have studied its application as an electrocatalyst for the oxidation and reduction of hydrogen peroxide, which would allow its use as a biosensor based on peroxidase enzymes. This novel application for this new HCN-derived polymeric film shows the huge unexplored potential of these complex conjugated C-N networks, whose properties are directly and markedly dependent on the reaction conditions used for their production [3].

Thus, in an effort to explain the electrochemical differences found between the DAMN polymer synthesized according to solid-state polymerization (SSP) (DAMN polymer-SSP) and under melt polymerization (MP) (DAMN polymer-MP), a detailed structural characterisation was carried out. This characterisation, together with their thermal and morphological properties, was based on elemental compositional analysis, Fourier transform infrared (FTIR), nuclear magnetic resonance (NMR) and X-ray photoelectron (XPS) spectroscopies, powder X-ray diffraction (XRD), scanning electron microscopy (SEM) and different thermoanalytical techniques such as thermogravimetry (TG) and differential scanning calorimetry (DSC). Hypothetical pathways for the formation of their macrostructure are comprehensively discussed below, with observations of tiny differences between the two polymers as a cause of their dissimilar electrochemical behaviour.

Interestingly, the films used for the modification of the electrodes in this study were synthetized only by thermal activation in the absence of solvent. In contrast, all the previous examples reported for HCN-derived coatings have been prepared in situ by direct immersion of the substrates in AMN solutions or cyanide aqueous media. Using this wet methodology all polymeric films contain oxygen in their structures [8,12] as well as insoluble black polymers produced by the water polymerization of DAMN, where hydrolysis and other oxidation processes must be taken into consideration in complex polymerization pathways [17]. However, the thermal activation of DAMN described here leads to HCN-derived macrostructures free of oxygen and presumably to extended conjugated two-dimensional (2-D) structures. The absence of oxygen in their macrostructure is relevant for the conductivity of polymers [18].

The results presented here offer the possibility of exploring HCN-derived polymers not only as important abiotic sources of biomolecules as they have been regarded in the past [19–23] but also as substances with a wide potential catalogue of properties such as other C-N networks and the widely studied graphitic carbon nitrides (g-C_3N_4) [24], also reported as available for biosensing applications [25]. In this way, DAMN polymers may significantly increase the prebiotic chemistry space as well as their possible applications in several areas of materials science. Moreover, the highly efficient thermal syntheses reported here for these bulk polymerizations present a promising industrial scaling factor since they are highly robust, easily handled, cheap, and solvent-free processes. Here it will be demonstrated that basic prebiotic chemistry simulation experiments under plausible early Earth environmental thermal conditions can lead to the generation of a successful and efficient nonenzymatic sensor, producible using a low-cost solvent-free process that can be carried out at an industrial scale, and underlining the promising future of HCN-derived polymers as multifunctional materials.

2. Materials and methods

2.1. Thermal syntheses of the DAMN polymers

Four hundred milligrams of DAMN was set into 20 mL sealed glass vials fitted with Teflon/silicone septa under an inert atmosphere of nitrogen and heated in heating blocks at 170 °C for 3 h (SSP) or at 190 °C for 20 min (MP). The temperature and reaction times were chosen based on previous works, as commented above, and on the DSC data analysis reported here, explained below. The final reaction products were black solids collected directly in the vials and characterised and measured without any type of additional handling.

At least three syntheses for each temperature were carried out to determine the robustness of the processes. The final weights of the DAMN polymers obtained were 352 ± 7.9 mg for the DAMN polymer-SSP and 350.4 ± 7.0 mg for the DAMN polymer-MP, with final yields of 88 ± 2% in both cases, considering the initial mg of DAMN [Conversion (%) = (mg of DAMN polymer/mg of DAMN) × 100]. The weight lost during the polymerization processes was due to deamination and decyanation, discussed below. DAMN (98%) was purchased from Sigma Aldrich and was used as received.

2.2. DSC measurements and thermal analysis

Isothermal runs were carried out in a Perkin Elmer DSC-7 calibrated with high-purity chemicals. Samples of approximately 10 mg of DAMN were placed in aluminium pans and isothermally polymerized at different temperatures under a 20 mL min^{-1} dry nitrogen atmosphere. To select suitable temperatures for the isothermal runs, dynamic scans were previously performed and described in recent works [15,16]. Temperatures above but near the onset of reaction were chosen. Experiments at 190 °C were conducted to investigate the DAMN MP, while two temperatures of 150 and 170 °C were applied when the SSP was analysed. In these measurements, the sample pan was stabilized at 30 °C, and then the temperature was raised to the programmed temperatures at a heating rate of 200 °C/min. This is to minimize the unrecorded reaction heat during the initial heating before data collection was initiated. The experiments were maintained at these temperatures until there was no change in the heat flow. Each test was repeated at least three times to ensure consistent results. The reaction heats were estimated by drawing a straight line connecting the baselines before and after the peak and integrating the area under the peak. The final values are the average of several measurements taken for each sample.

Thermal degradation properties of the DAMN polymers were studied using TG, DSC and differential thermal analysis (DTA) measurements with an SDTQ-600/Thermo Star simultaneous thermal analyser (TA Instrument). The nonisothermal experiments were carried out under dynamic conditions from room temperature to 1000 °C at a heating rate of 10 °C min^{-1} under an argon atmosphere. The average sample weight was ~ 20 mg, and the argon flow rate was 100 mL min^{-1}. A coupled TG-mass spectrometer (TG-MS) system equipped with an electron-impact quadrupole mass-selective detector (model Thermostar QMS200 M3) was employed to analyse the main species evolved during the dynamic thermal decomposition of all of the samples. This same technique was applied to identify the gaseous species generated during the DAMN isothermal polymerizations at 190 °C.

2.3. Electrochemical measurements

The electrochemical studies were performed using an Ecochemie BV Autolab PGSTAT 12 in a conventional three-electrode cell at 20–21 °C with a Pt wire as the auxiliary electrode and a saturated calomel reference (SCE) electrode. The working electrodes were prepared by dropping 5 μL of a sonicated suspension (1 mL of absolute ethanol and 5 mg of the corresponding material, with a sonication time of 15 min) on the surface of Pt disc (3 mm diameter) electrodes and drying them in air at room temperature. Before the measurements, the modified electrodes were activated with a potentiostatic treatment at −1.0 V for 30 s to remove material not adhered to the electrode surface and to facilitate the circulation of ions through the film. Electrochemical impedance spectroscopy (EIS) measurements were carried out in a 10 mM $K_3Fe(CN)_6/K_4Fe(CN)_6$ 1:1 solution containing 0.1 M KCl, at 200 mV *vs.* Ag/AgCl electrode, over the frequency range of 0.1 to 10.000 Hz, with 10 mV AC perturbation.

2.4. Structural characterisation and morphological features

2.4.1. Elemental analysis

The DAMN polymers were examined to determine the mass fractions of carbon, hydrogen and nitrogen using a Perkin Elmer elemental analyser, model CNHS-2400. The percentages of oxygen in the samples were obtained by calculating the differences.

For the DAMN polymer-MP, the following data were obtained: %C = 42.15 ± 0.34; %H = 3.40 ± 0.02; %N = 48.66 ± 0.48; %O = 2.79 ± 0.83. For the DAMN polymer-SSP, %C = 45.20 ± 0.04; %H = 3.32 ± 0.19; %N = 48.83 ± 0.67; %O = 2.64 ± 0.90. The macrostructures of both polymers are free of oxygen, and their low percentage is due to moisture due to the high hydrophilicity of the samples, as confirmed by thermal analysis measurements.

2.4.2. FTIR spectroscopy

Diffuse reflectance spectra were acquired in the 4000–400 cm^{-1} spectral region using an FTIR spectrometer (Nicolet, model NEXUS 670) configured with a drift reflectance accessory (Harrick, model Praying Mantis DRP) mounted inside the instrument compartment. The spectra were obtained in CsI pellets, and the spectral resolution was 2 cm^{-1}.

2.4.3. ^{13}C solid-state cross polarization/magic angle spinning (CP/MAS) NMR spectroscopy

^{13}C CP/MAS NMR spectra were obtained using a Bruker Advance 400 spectrometer and a standard cross-polarization pulse sequence. Samples were spun at 10 kHz and the spectrometer frequency was set to 100.62 MHz. A contact time of 1 ms and a period between successive accumulations of 5 s were used. The number of scans was 5000 and the chemical shift values were referenced to tetramethylsilane (TMS).

2.4.4. XPS spectroscopy analysis

X-ray photoelectron spectroscopy analysis of samples was carried out in an ultrahigh vacuum chamber equipped with a Phoibos 150 MCD spectrometer with a hemispherical electron analyser using an Al Kα X-ray source (1486.7 eV) with an aperture of 7 mm × 20 mm. The base pressure in the chamber was 3×10^{-8} mbar and the experiments were performed at room temperature. A 30 eV pass energy was applied for taking the overview sample, where a 20 eV pass energy was applied for the analysis of the following core level spectra: C 1s, O 1s and N 1s. The peak analysis in different components was shaped, after background subtraction, as a convolution of Lorenztian and Gaussian curves. Binding energies were calibrated against the binding energy of the C 1s peak at 285.0 eV. XPS spectra regions were fitted and deconvoluted using fitt-xps software. The samples were analysed from a pellet of the sample prepared by slightly pressing the powder materials.

2.4.5. XRD

Powder XRD measurements were performed using a Bruker D8 Eco Advance with Cu Kα radiation (λ = 1.542 Å) and a Lynxeye XE-T linear detector. The X-ray generator was set to an acceleration voltage of 40 kV and a filament emission of 25 mA. Samples were scanned between 5° (2θ) and 60° (2θ) using a step size of 0.05° and a count time of 1 s using the Bragg–Brentano geometry.

2.4.6. SEM

The surface morphology of the DAMN polymeric particles was assessed using a Thermo Scientific Apreo C-LV field emission scanning electron microscope (FE-SEM) equipped with an Aztec Oxford energy-dispersive X-ray microanalysis system (EDX). The samples were dispersed by sonication in EtOH, deposited on glass slides, dried in air and then coated with 4 nm chromium by sputtering using a Leica EM ACE 600. The images were obtained at 10 kV.

3. Results and discussion

3.1. Self-initiated thermal polymerizations

In previous studies the peculiar thermal behaviour of DAMN was described from dynamic DSC experiments, where it was clearly evidenced that its polymerization reaction occurs in the solid-state before the onset of the melting process and after this transition. However, this dual behaviour is not observed when high heating rates are imposed, and only the MP occurs at temperatures above 180 °C [16]. On the other hand, DSC scans at very low heating rates allow us to address the SSP, but all polymerizations did not reach completion [15]. These dynamic measurements hinted at the choice of reaction temperature; thus, we designed isothermal experiments at 190 °C to study MP, and thermal polymerizations at 150 and 170 °C were performed to evaluate the polymerization in the solid-state.

Fig. 1 shows two DSC grams for 170 and 190 °C describing the entire programmed experiments. The initial data indicated that a very short time is required to reach the selected polymerization temperature. In the case of 190 °C the melting peak can be clearly observed before the start of the reaction. As seen from these thermograms, the appearance of a maximum at time different from zero provides evidence of the existence of an overall autocatalytic reaction mechanism, as was concluded when both polymerization processes were analysed under nonisothermal conditions [15,16]. The isothermal heat flow curve as a function of time for the MP showed a highly exothermic character, and at the same time this reaction was very fast, fully completed in a few minutes. In addition, through integration of the heat flow curves under the baseline the heat generated during the MPs was determined, and taking as reference the polymerization enthalpy of 813 J g^{-1} (or 87.8 kJ mol^{-1}), obtained from traditional dynamic mode, monomer conversions can be estimated [16]. The measured reaction heats for isothermal samples were acceptably close to this value and were thus assumed to be polymerized completely. This fact was confirmed by FTIR analyses, discussed later. The characteristic parameters describing the heat flow time dependencies are presented in Table 1 and mostly represent the average of several measurements.

The other inset plot in Fig. 1 illustrates the DSC runs for the two temperatures under study for the SSP (150 °C and 170 °C). It is possible to observe that this occurs over a longer period of reaction time, and at the same time the polymerization heat is remarkably lower than in melt, confirmed when dynamic experiments were conducted. From these nonisothermal DSC data a value of 520 J g^{-1} for the SSP enthalpy was established [15]. As the reaction temperature decreases the exothermic

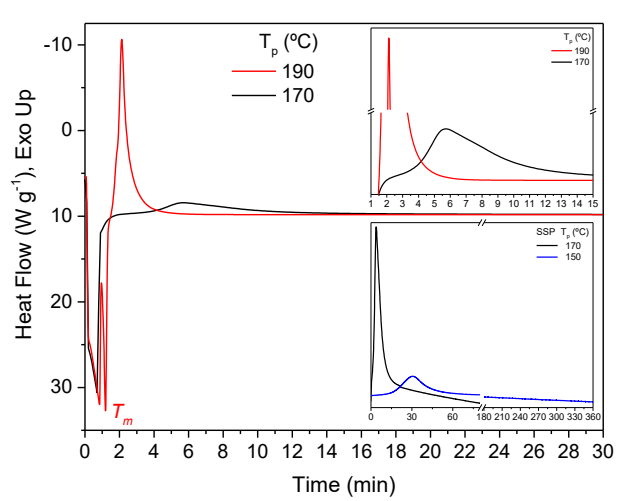


Fig. 1. DSC curves for the DAMN isothermal bulk polymerizations under nitrogen atmosphere.

Table 1
Main parameters of the DSC thermograms for DAMN isothermal bulk polymerizations. *DSC conversion considering the amount released heat during the thermal processes, not necessarily DAMN polymers formed.

T_p (°C)	Peak Time (min)	Released Heat (J g^{-1})	Conversion (%)*
190	3.2	809	99.5
170	6.8	517	99.4
150	31	445	85.6

area is shifted to higher reaction times, so the DSC peak at 170 °C appears after 7 min of isothermal polymerization, while at the lowest temperature under study the maximum polymerization rate takes place at approximately 30 min. It is important to note that the final conversion was reached only for SSPs at 170 °C. This final conversion must be understood as the full consumption of DAMN and not as a yield of 100% in the polymerization process, as seen below. In contrast, the heat flow ceased at 150 °C after 6 h of polymerization, giving a conversion degree of 86%. The FTIR spectrum of this sample showed a signal of residual DAMN, confirming the kinetic results of the SSP at this temperature. For this reason it was not subject to further analyses.

These results allowed us to carry out preparative syntheses at 170 and 190 °C for the reaction times defined by DSC measurements. Gravimetry, as described in Section 2.1, revealed that approximately 12% of weight was lost in both series. This result was in agreement with a previous report by Mamajanov and Herzfeld [26] and is strongly supported by thermogravimetric analyses which revealed that the dynamic thermal polymerization of DAMN is accompanied by the elimination of HCN and NH_3 [15,16]. This volatile elimination is also observed under isothermal conditions, as will be discussed below.

3.2. Electrochemical characterisation and applications

Following one trial in which the DAMN polymers were synthetized properly, their potential as conductive polymers and electrocatalytic materials was explored. Thus, electrochemical characterisation of electrodes modified with them was carried out. Fig. 2 shows the cyclic voltammograms (CVs) of modified electrodes with both DAMN polymers in aqueous $NaClO_4$ 0.1 M. Both materials present similar CVs, with two quasi-reversible electrochemical systems at very close potentials (around 0.0 and −0.6 V respectively), indicating the same electroactive functional groups in both polymers, although with a different quantitative relationship between them. System 1 of DAMN polymer-MP shows

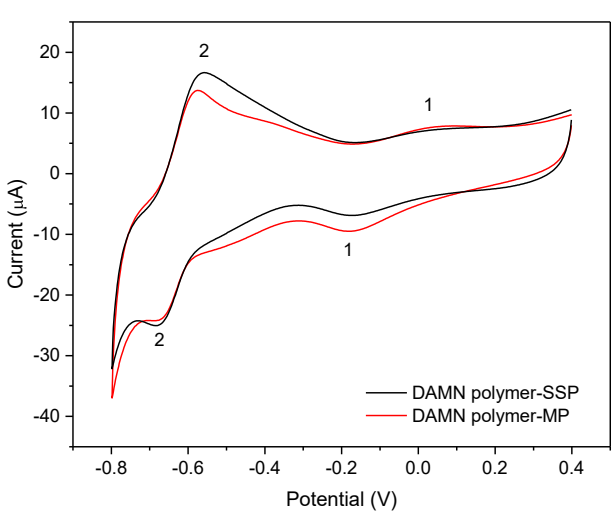


Fig. 2. CVs of electrodes modified with DAMN polymer-SSP (black line) and DAMN polymer-MP (red line) measured in 0.1 M $NaClO_4$ at a scan rate of 100 mV s^{-1}. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a higher current peak than that of DAMN polymer-SSP, while the peaks of system 2 of this sample are more intense than that of DAMN polymer-MP.

These two systems have been characterized from the slopes of the Tafel plots (log *i vs.* overpotential) [27] (Figs. S1 to S4), which allow us to know the number of electrons exchanged, n, the electron transfer coefficient, α, and the exchange current density, j_0. The results collected in Table 2 show that in all cases the redox systems exchange one electron, and their symmetries are similar regarding the α values, indicating higher symmetry of systems 1 with respect to 2 for both types of electrodes. On the other hand, j_0, equal in magnitude to either component current, j_c or j_a, when the net current is zero at equilibrium and indicative of reversibility of the redox systems, indicates less reversibility for systems 1 (lower j_0) than 2 in both cases.

Study of the kinetics of modified electrodes with mono- or multilayers strongly adsorbed on the electrode surface is usually carried out using the model developed by Laviron [28,29] based on the variation of the current–potential curves with the scan rate (v) (Fig. 3). Plots of the anodic and cathodic peak currents *vs.* the scan rate allow us to estimate if the electroactive groups are surface confined. If this occurs, the plots must show a linear relationship in some scan rate intervals. In our case, both systems of each DAMN polymer show linear dependence at high sweep rates (Fig. S5), confirming the surface confined nature of the electroactive groups. In addition, the proportionality ip-v at high scan rates indicates the exhaustive oxidation–reduction of the redox sites through the electrode coverage during the potential sweep [30]. However, Laviron's treatment demonstrated that the anodic and cathodic peak currents only show linear behaviour when v is large or small, but in the middle or the whole range ip will be proportional to v^x, taking x values between 0.6 in the case of multilayer films with fast electron transfer and the unit (nonexistence of charge percolation in a monolayer film). As shown in the bottom insets of Fig. 3, all the studied systems present a good linear relation between the anodic or cathodic peak currents and $v^{0.6}$ over the whole sweep rate interval and consequently, in agreement with Laviron's model, we can assert that the modified electrodes have a multilayer structure with fast electron transfer [29].

On the other hand, the variation of the peak potentials with the scan rate can also provide relevant information on the kinetics of the modified electrode. The graphs of peak potential *vs.* the logarithm of the scan rate are increasing, yielding straight lines, indicating some kinetics limitation to the polymer-electrode electron transfer. The study of this variation allows us to estimate the homogeneous standard rate constant k_s for the electron transfer electrode-confined redox system. For this study, Laviron's model distinguishes two cases as a function of the magnitude of the separation of the anodic and cathode peak potentials, ΔE_p [29]. For systems and v values with $\Delta E_p > 200/n$ mV, k_s can be calculated using the equation:

$$log\, k_s = alog(1-a) + (1-a)loga - log(RT/nFv) - a(1-a)nFDEp/2.3RT \quad (1)$$

where R is the gas constant, F is the Faraday constant and T is the absolute temperature. If the experimental conditions are such that only values of $\Delta E_p < 200/n$ mV can be obtained, the model offers a table of values of ΔE_p as a function of $1/m$, where $m = RTk_s/Fnv$, and the polynomial fit ($R^2 = 0.9992$) is:

Table 2
Results of the electrochemical characterisation of electrodes modified with the polymers synthesized from DAMN isothermal bulk polymerizations.

Redox System	$E^{0\prime}$ (V)	n	α	k_s (s^{-1})	J_0 (A cm^{-2})
DAMN polymer-MS-1	0.024	1	0.62	0.14	$3.17 \cdot 10^{-7}$
DAMN polymer-MS-2	−0.61	1	0.72	2.86	$8.37 \cdot 10^{-5}$
DAMN polymer-SSP-1	−0.095	1	0.55	0.05	$9.26 \cdot 10^{-7}$
DAMN polymer-SSP-2	−0.62	1	0.61	1.80	$1.41 \cdot 10^{-4}$

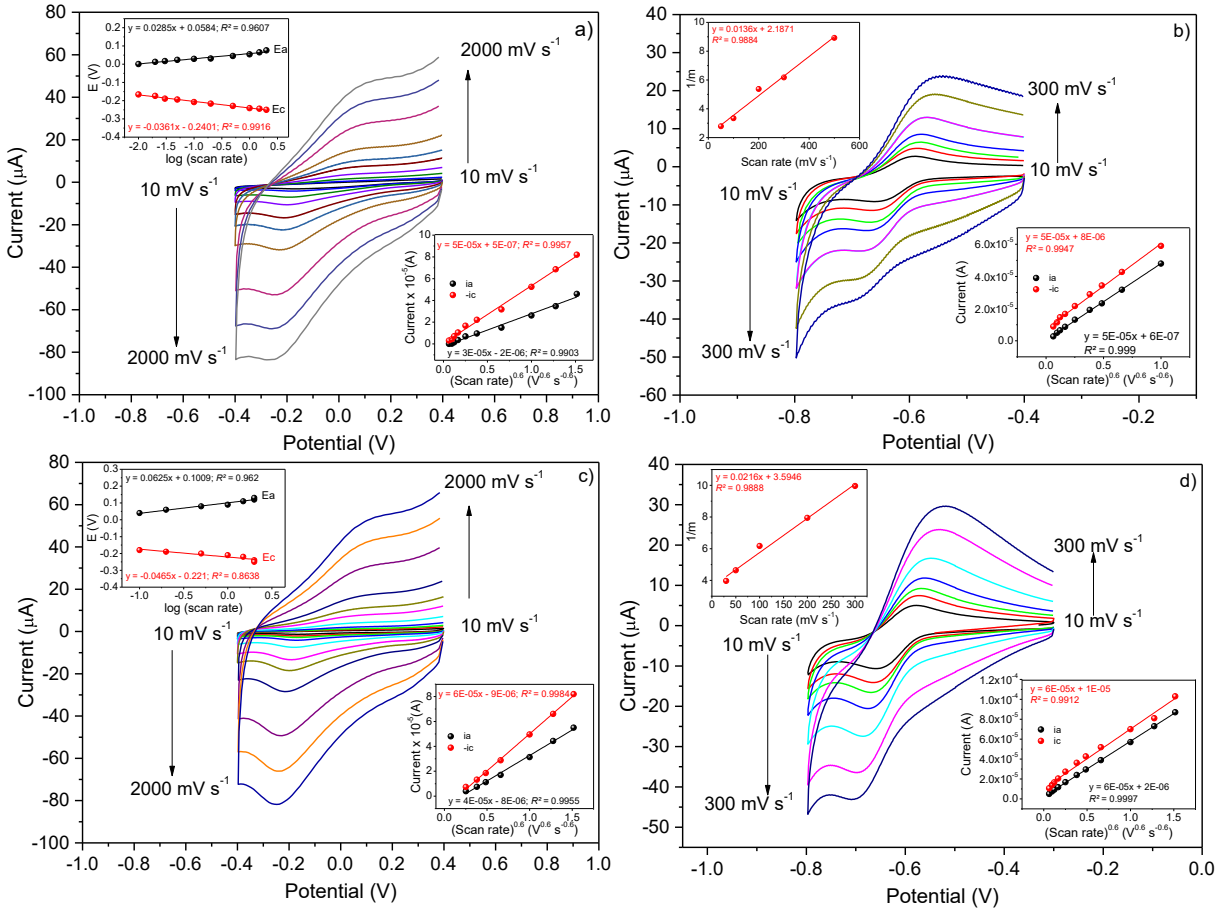


Fig. 3. Dependence of the current–potential curves on the scan rate in 0.1 M $NaClO_4$ of (a) system 1 of DAMN polymer-MP, (b) system 2 of DAMN polymer-MP, (c) system 1 of DAMN polymer-SSP and (d) system 2 of DAMN polymer-SSP. Upper inset plots: (a) and (c) are corresponding plots of the variation of anodic and cathodic potential with the logarithm of the scan rate, and (b) and (d) are corresponding plots of 1/m *vs.* scan rate. Bottom inset plots: corresponding plots of the dependence of anodic and cathodic peak currents on the (scan rate)$^{0.6}$.

$$\frac{1}{m} = \frac{Fnv}{RTk_s} = 0.0003\Delta E_p^2 + 0.0047\Delta E_p + 0.491 \quad (2)$$

In our case, systems 1 of both polymers shows $\Delta E_p > 200/n$ mV (upper insets in Fig. 3**a** and **c**), while systems 2 shows $\Delta E_p < 200/n$ mV (the upper insets in Fig. 3**b** and **d** show the plots of $1/m$ *vs.* v); therefore, we must use both models to calculate k_s. The results are included in Table 2. From these results, we can conclude that systems 2 are faster than 1 and that the DAMN polymer-MP sample shows faster kinetics than the DAMN polymer-SSP.

EIS studies allowed us to know the properties of the electrode interface. The EIS technique allows modelling of the electrode interface using an equivalent circuit composed of the charge-transfer resistance R_{CT}, which controls the electron transfer kinetics, the Warburg impedance, W (diffusion of ions from the bulk electrolyte to the electrode interface), the interfacial (double layer) capacitance, C_{dl}, and the electrolyte resistance, R_s. Fig. 4 displays the Nyquist plots obtained with both polymer sample-modified electrodes and bare Pt, including the modified Randles equivalent circuit selected to fit the measured results. The Nyquist plots consist of a semicircular part whose diameter represents R_{CT} and a linear part at low frequencies characteristic of systems with diffusion-controlled current. The R_{CT} shown by the DAMN polymer-MP-modified electrode (70.54 Ω) is only slightly higher than that of Pt bare (35.694 Ω), while the sample produced by the DAMN polymer-SSP-modified electrode presents a higher value (131.44 Ω). Nevertheless, these values are very low and indicate good conductivity, although the DAMN polymer-MP shows better conducting properties. The interfacial capacitance was similar in all cases. Values of 0.85, 0.72 and 0.99 μF were obtained for Pt bare, DAMN polymer-MP and DAMN polymer-SSP modified electrodes, respectively.

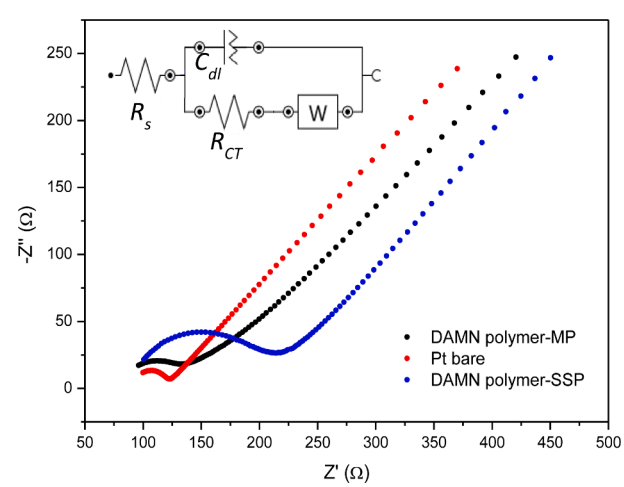


Fig. 4. Electrochemical impedance spectra and equivalent circuit of DAMN polymer-SSP- and DAMN polymer-MP-modified electrodes and Pt bare. All measurements in a solution of 0.1 KCl with 10 mM $Fe(CN)_6^{4-/3-}$.

In view of the good electrochemical, kinetic and conductive properties of these polymers, according to the Mamajanov and Herzfeld study where in that case the DAMN polymer was doped with iodine [26],

their application as electrocatalysts for the oxidation and reduction of hydrogen peroxide has been studied. We selected hydrogen peroxide because of its importance in the development of electrochemical biosensors that include oxidase enzymes. Hydrogen peroxide is produced in the biochemical reactions catalysed by these enzymes during the oxidation of the corresponding substrates, and their determination is a way of indirect determination of substrates such as glucose [31] or xanthine [32]. In addition, hydrogen peroxide degradation can be considered a potential method for the identification of bioorganics in extraterrestrial bodies [33], and from a prebiotic chemistry point of view H_2O_2 decomposition can be a significant biological redox reaction useful for studying the possible catalytic activity of protoenzymes [34]. The major limitations for peroxide determination are the high operating potential required for peroxide oxidation and the overlap of the reduction signal with those of the reduction of oxygen necessarily present in the enzymatic reaction for the production of peroxide. For these reasons, the goal of new development is to reduce overvoltage, minimize interferences, and improve the electrochemical sensitivity [35].

Fig. 5 shows the typical current–potential CVs for oxidation and reduction of H_2O_2 at bare and modified electrodes. Both oxidation and reduction of peroxide take place at more extreme potentials in the case of bare Pt than on modified electrodes, indicating that DAMN polymer-MP acts as an electrocatalyst for both reactions. The DAMN polymer-SSP only shows current enhancement for the reduction reaction. The current at the corresponding peak potential was enhanced by 10619% (DAMN polymer-MP at −0.1 V), 247% (DAMN polymer-MP at 0.2 V) and 8942% (DAMN polymer- SSP at −0.5 V) in relation with the Pt bare electrode. These increases indicate that DAMN polymer-MP is the best electrocatalyst and that catalysis is more effective in the reduction rather than the oxidation reaction and are in agreement with previous studies.

Next, a kinetic study of the reactions of interest was carried out with the modified electrodes by means of a rotating disk electrode. The Levich plots (i_{lim} *vs.* $\omega^{1/2}$) of modified electrodes (not shown) at low rotation rates are linear, indicating mass transport-limited processes, and at higher rotation rates these plots deviate from linearity showing kinetic limitations. In these cases, the Koutecky-Levich equation [36,37] can be used to obtain the kinetic heterogeneous rate constants, k:

$$\frac{1}{i} = \frac{1}{nFAkC_0} - \frac{1}{0.620nFAD_0^{2/3}\nu^{-1/6}C_0\omega^{1/2}} \quad (3)$$

where n is the number of electrons (2 in all cases), F *is* the Faraday constant, A is the electrode area, ν *is* the kinematic viscosity, D_0 is the diffusion coefficient, ω is the angular velocity, and C_0 is the bulk concentration of H_2O_2. The diffusion coefficients calculated from the slopes of Koutecky-Levich plots (i^{-1} *vs.* $\omega^{-1/2}$) (Fig. S6) and the rate constants calculated from the intercepts are collected in Table 3. The obtained D_0 values are similar in all cases and somewhat lower than the value in solution ($1.71 \cdot 10^{-5}$ cm^2 s^{-1}) [38] due to percolation into the polymer film, indicative that the overall reaction rate is diffusion-controlled. The obtained k values were dependent on peroxide concentration (Fig. S6), so their values must be estimated as $k_{[H2O2]=0}$ from the fit of the plot k *vs.* concentration. This fact has been observed previously for catalytic NADH oxidation and for hydrogen peroxide reduction [35,39], and has been explained by the formation of a charge transfer complex prior to product formation, following the reaction scheme:

I- $H_2O_{2\ bulk} \rightarrow H_2O_{2\ surface}$

II- $H_2O_{2\ surface} + H^+ + e^- \rightarrow OH_{ads} + H_2O$

III- $H_2O_2 + H^+ + OH_{ads} + e^- \rightarrow 2OH_{ads} + H_2O$

According to this sequence, reactions II and III reach the formation of OH_{ads} and this adsorbate increases the reaction rate. We can conclude that the reduction of peroxide proceeds by an autocatalytic mechanism. Likewise, the occurrence of adsorbed intermediate species OH_{ads} has been advanced to explain a pseudolimiting current and the relatively small current density observed in the oxidation reaction [40]. Regarding the values of the constants, they confirm the higher efficiency of the DAMN polymer-MP with respect to the DAMN polymer-SSP and the most efficient electrocatalysis of the reduction reaction.

The overall results indicate that the DAMN polymer-MP is the best electrocatalyst for hydrogen peroxide reactions, mainly for the reduction reaction. Consequently, we selected this polymer to carry out the analytical study. Fig. 6 shows the calibration plot obtained with the mean values of five replications at the DAMN polymer-MP-modified electrode at −0.3 V working potential. The amperometric steady-state response time was < 1 s (inset plot), demonstrating the electrocatalytic effectiveness of the modified electrode. From this figure, we can observe that the signal varies linearly with the concentration of peroxide at least up to 1.5 mM, with a high sensitivity of 485.7 ± 3.3 μA

Table 3
Kinetic and analytical average results ($n = 4$) for the reduction and oxidation of hydrogen peroxide.

Electrode	Applied Potential (V)	D_0 (cm^2 s^{-1})	k (cm s^{-1})
DAMN polymer-MS	+0.5	$0.88 \cdot 10^{-5}$	0.039
	−0.3	$0.88 \cdot 10^{-5}$	1.69
DAMN polymer-SSP	−0.3	$0.90 \cdot 10^{-5}$	0.017

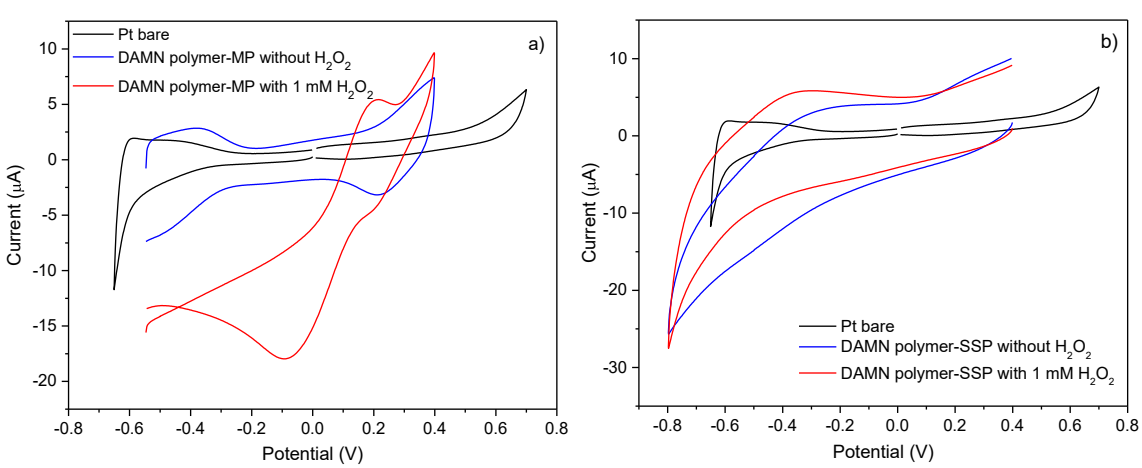


Fig. 5. CVs of (a) DAMN polymer-MP and (b) DAMN polymer-SSP-modified electrodes in the absence and presence of 1 mM hydrogen peroxide in deaerated 0.1 M $NaClO_4$ supporting electrolyte. The black anodic and cathodic CVs correspond to a Pt bare electrode in the presence of hydrogen peroxide 1 mM in the same supporting electrolyte. Scan rate 50 mV s^{-1}.

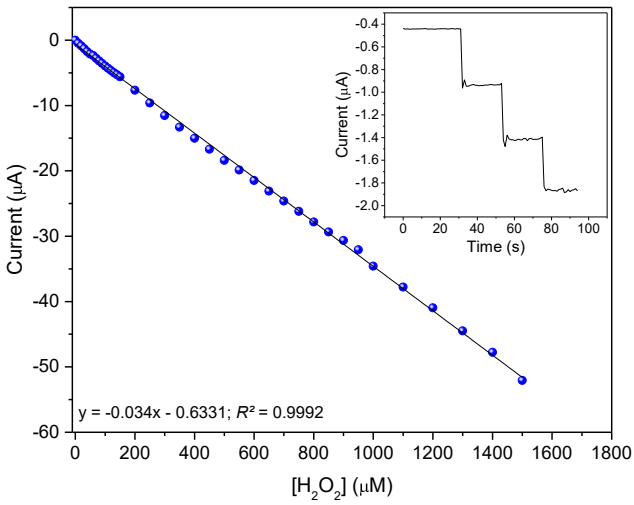


Fig. 6. Cathodic hydrogen peroxide calibration plot obtained at Pt electrodes modified with DAMN polymer-MP at −0.3 V *vs.* SCE. Inset plot: amperometric response of a rotating modified electrode to successive additions of aliquots of 10 μM H_2O_2. All measurements were in deaerated pH 7, 0.01 M phosphate buffer/0.1 M $NaClO_4$ supporting electrolyte and at 500 rpm.

$mM^{-1}cm^{-2}$ and limit of detection of 0.11 μM calculated as twice the standard deviation of signal/slope of the calibration lines.

To consider applying the new electrode as a nonenzymatic peroxide sensor as well as for catalytic support of oxidase enzymes, the selectivity was studied. The usual interfering substances, uric acid, ascorbic acid, glucose, and others, do not interfere at cathodic potentials. The presence of oxygen did not cause any appreciable signal at the measuring potential (-0.3 V) in the presence of peroxide, so it can be considered that the sensor is interference-free. Finally, the reproducibility of the electrode was examined in a solution of 50 μM peroxide, obtaining a relative standard deviation of 3.2% (n = 5) and a relative standard deviation electrode-to-electrode of 5.3%. The stability of electrodes was evaluated for 15 days with intermittent measurements of peroxide 50 μM, and their response remained unchanged during the full period.

These results demonstrate that the DAMN polymer-MP-modified electrode is very competitive, even in some cases better than other more compact electrodes recently developed [35].

3.3. Structural study and morphological characteristics

Due to the dissimilar electrochemical behaviour observed between the DAMN polymers, a detailed structural characterisation is necessary to understand this difference from the perspective of the relationship structure-properties. Note that although the DAMN polymers could modify the electrodes after sonication in EtOH they present a very low solubility in all common organic solvents. Thus, their molecular characterisation was totally conditioned by this fact.

The different polymeric samples were handled under air at atmospheric pressure and evaluated by elemental analysis. The molar C/H relationships found were 1.136 ± 0.065 and 1.108 ± 0.003 for the DAMN polymer-SSP and DAMN polymer-MP, respectively. The values obtained for the molar C/N ratios were 1.080 ± 0.014 and 1.083 ± 0.002 for the DAMN polymer-SSP and DAMN polymer-MP, respectively.

The FTIR spectra present the same features reported recently in nonisothermal kinetic studies for analogous DAMN polymers (Fig. 7**a**). Therefore, a rigorous interpretation of these spectra herein will not be considered because they were made elsewhere [15,16]. However, reviewing the nitrile region in detail in the 2270–2170 cm^{-1} range, two consecutive peaks, at ~ 2210 and ~ 2170 cm^{-1}, have been found (inset plot of Fig. 7**a**) which have been described previously for other HCN-derived polymers [41]. Both features can be related to nitrile groups conjugated with a double bond being likely part of the heteroaromatic ring [42].

Further quantitative data for describing the different behaviour of the two polymerizations can be obtained from the intensities between the following peaks at ~ 3345 cm^{-1} (amine groups), I_{3345}, the nitrile peak at ~ 2195 cm^{-1}, I_{CN}, and ~ 1625 cm^{-1} [C = C and/or $(C = N)_n$], I_{1625}, [17,43]. Comparatively, no significant differences were found for the DAMN polymer-SSP $I_{1625}/(I_{2195} + I_{1625})\cdot 100 = 59.3 \pm 2.7$ and $I_{1625}/$

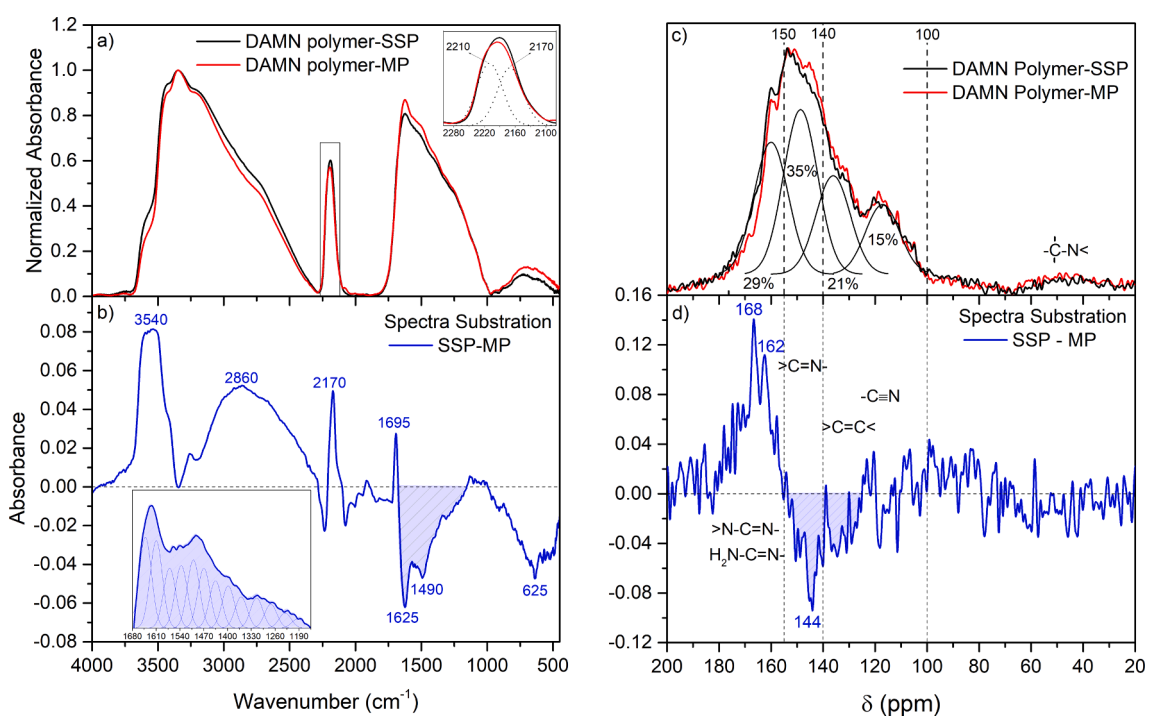


Fig. 7. a) Representative normalized FTIR spectra of DAMN polymers synthesized by SSP and MP; b) Subtraction of the FTIR spectra; c) ^{13}C NMR spectra; d) Subtraction of the NMR spectra.

$(I_{3345} + I_{1625})\cdot 100 = 45.9 \pm 1.7$ and 61.0 ± 0.9 and 47.2 ± 1.0, respectively, for the DAMN polymer-MP. Only subtle vibrational differences between the two samples were found when spectral subtraction was carried out (Fig. 7**b**). The subtraction shows a slight increase in absorption in the $-NH_2$ stretching region between 3750 and 3350 cm^{-1}; therefore, a greater number of amine groups should be present in the polymer samples obtained in the solid-state, and fewer deamination processes must occur under these milder thermal conditions. In addition, the positive peaks at 2170 and 1695 cm^{-1} confirm the presence of a β-enamino nitrile and/or β-imino nitrile corresponding to a nitrile moiety in conjugation with a double bond according with the literature data [42]. Finally, negative bands are found in the broad region between 1680 and 1190 cm^{-1}, where the typical stretching modes of CN heterocycles appear, demonstrating the formation of extended networks of C-N-C bonds, with relevant peaks at 1625 and 1490 cm^{-1} as illustrated in the inset plot [44]. From this last result, we think that more cyclization reactions could take place when the polymerization reaction is performed in the melt, as has been reported in the literature [45].

To find other structural differences beyond those described above, the ^{13}C CP/MAS NMR spectra of both samples are displayed in Fig. 7**c**. Both spectra presented the same broad features as described by Mamajanov and Herzfeld [26]. An explanation of this finding is that a repeated 2-D structural unit probably exists along with significant covalent cross-linking, which constrains motions and introduces structural heterogeneity. From the ^{13}C spectra, we think that the broad unresolved resonance at 60–100 ppm allows largely ruling out the presence of sp^3 carbons (C-N); however, different sp^2 carbon regions and a broad nitrile resonance can be identified. However, the deconvolution of four relevant resonances of these spectra could help to better interpret this complex and heterogeneous system. Thus, the strong featureless resonances at 180–135 ppm are from carbons in the carbon–nitrogen double bond functionality (C = N). The breadth of this imine region ($R_1R_2C = N\text{-}R_3$) may be due to the heterogeneity in R. Resonances above 150 ppm can be assigned to carbons bound to two or three nitrogens, such as $\text{-}(NH_2)C = N\text{-}$ [46,47]. At lower chemical shifts a distribution of sp^2 carbons connected with a single nitrogen atom is expected. The contributions of the corresponding Gaussian peaks shown in this figure are 29 and 35%, respectively. Therefore, approximately 64% of imine groups must be the major component of this polymeric system. Finally, the signals at 135–100 ppm can be assigned to carbons in the carbon–carbon double bond (C = C) (21%) or the carbon–nitrogen triple bond (C≡N) functionality (15%) (Fig. 7**c**) [26,46,47]. The percentage of nitrile groups over the macrostructure of the DAMN polymers seems relatively low considering the quantitative parameters obtained from the FTIR analyses.

In addition, following the same analytical methodology applied previously, spectral subtraction of NMR data can provide additional information and facilitate assignments. Again, only slight differences were found between the spectra of both samples (Fig. 7**d**). The positive peaks at 168 and 164 ppm suggest that $\text{-}N = C\text{-}NH_2$ fragments must be present to a greater extent in the sample obtained in the solid-state state. In contrast, and as extracted from the FTIR analysis, the highest percentage of heterocyclic structures in the sample produced from the melt must distribute their corresponding chemical shift in the region 110–155 ppm, with a stronger negative peak at 144 ppm as illustrated in this figure.

To obtain a deeper quantification of the functional groups and to resolve the apparent divergence between the FTIR and NMR results, especially the percentage of nitrile groups, XPS spectra were recorded. Fig. 8 shows the XPS core-level spectra of N1s and C1s peaks and their deconvolution on components of both samples. The N1s peaks of the samples were resolved in three components. The first peak at 399.0 eV

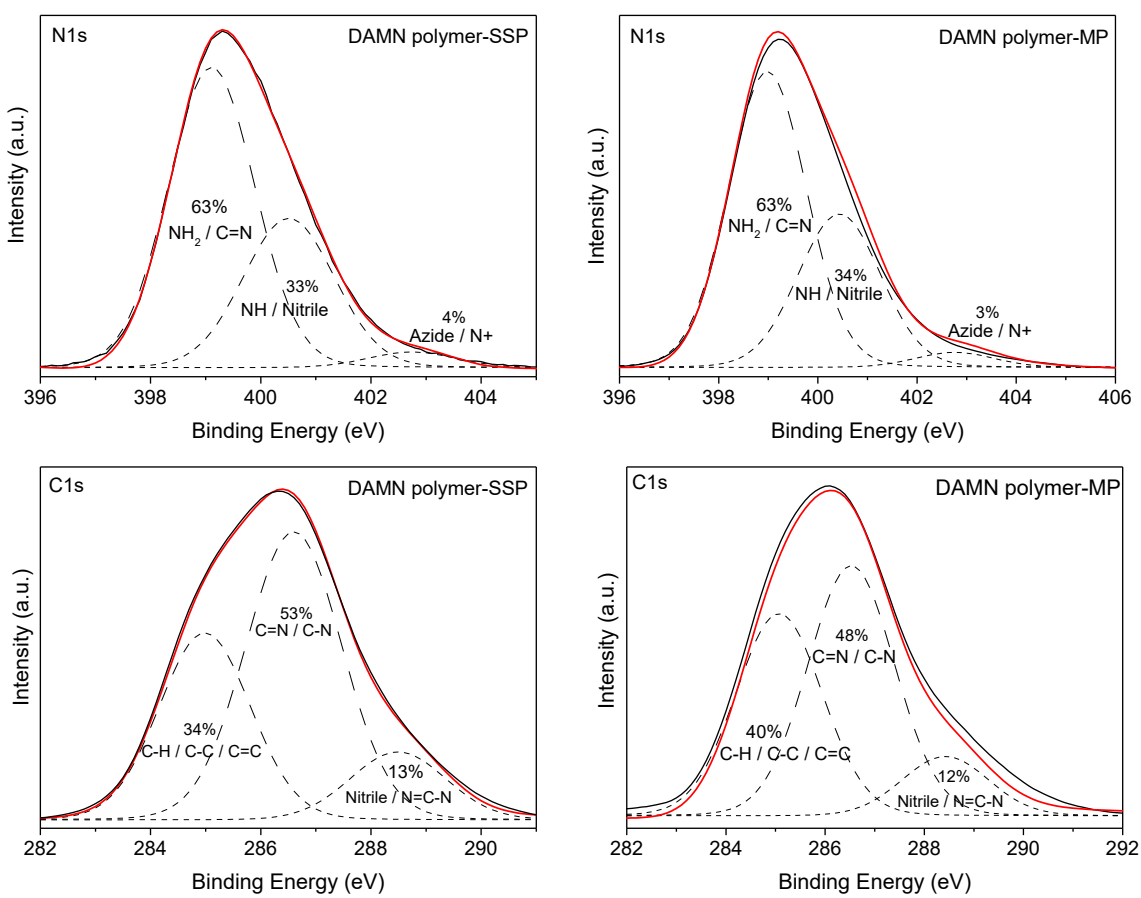


Fig. 8. XPS photoemission spectra of the N1s and C1s core level peaks of DAMN polymers synthesized by SSP and MP.

was assigned to amines (NH_2) and imines (N = C), the second peak at 400.5 eV corresponded to amines (NH) and nitriles, and the last peak at 402.7 eV can be related to azide chemical structures (R-N = N^+=N^-) or graphitic nitrogen [48–50]. The C1(s) XPS core level peak of the samples under study shows three components at 285.0 eV, 286.5 eV and 288.5 eV, the first assigned to C-H, C-C and C = C groups, the second to C = N/C-N, and the third to nitriles and N = C-N [51]. Both samples show similar ratio components, as indicated in Fig. 8. Therefore, there is good consistency between the data obtained by NMR and XPS, indicating that the main frame of the DAMN polymeric macrostructures is based on C = N bonds in which there are approximately 12–15% nitrile groups.

Finally, the XRD profiles suggest an approximately planar arrangement (Fig. 9**a**). Different studies have observed that HCN-derived polymers exhibit XRD patterns that contain only a few broad features [17,43,52]. They are typically dominated by a main peak at approximately 2θ 27° (Cu K_α radiation), which has been interpreted as an indication of the presence of a possible graphitic structure. Both DAMN polymers exhibited a stronger diffraction at ~ 26.5° related to (0 0 2), a d-spacing of ~ 3.25 Å, and a weaker reflection at 13° (1 0 0) with the same interlayer distance. Similar XRD patterns were observed for layered carbon nitrides [14,24,53]. The crystallinity percentage for the DAMN polymer-SSP was 56% and for the DAMN polymer-MP was 66%. Both values are higher than those reported for DAMN polymers synthesized in water solutions [17]. Note that the crystallinity of the systems increases with the reaction temperature, as expected [43,52].

In addition, knowledge about the morphology and textural properties of our C-N systems is of interest considering that these properties could be conveniently modulated for further applications [54] and to understand the differences produced by the SSP and MP processes. Thus, morphological investigation of the prepared samples was performed with SEM (Fig. 9**b** and **c**). In general, both DAMN polymers present no well-defined particle shapes.

As a result of the above characterisations the main structural differences found between the DAMN polymers are that the DAMN polymer-MP seems to contain a greater amount of N-heterocycles in its macrostructure and its crystalline fraction is higher than that of the DAMN polymer-SSP.

3.4. Thermal decomposition properties

Derivative thermogravimetry (DTG) and DSC curves have been used previously as thermal fingerprints of HCN-derived polymers when FTIR spectra are not capable of distinguishing samples synthesized under different conditions [55–58], such as in the present case, where other detected spectroscopic differences are certainly few. Fig. 10**a** shows the TG and DTG curves for both DAMN polymers. According to our preceding studies [55–58], the thermal degradation of these samples was divided into three stages: drying (<225 °C, stage I, elimination of volatiles, such as H_2O, with a mass loss of ~ 4%), main decomposition (225–450 °C, stage II, thermal decomposition of the weakest bonds, with a mass loss of ~ 9%) and carbonization (>450 °C, stage III, break along the main chain, with a mass loss of ~ 64%), leading to a char residue approximately 24% in weight.

As DTG curves are more sensitive than TG-grams they are shown in the same figure on the right side. Slight differences are perceived in the thermal behaviour of the samples, mainly in stages I and III. On one hand, in stage I variances are found between both series, as indicated in the inset plot, where a deconvolution to establish the number of distinct volatile compounds is presented. Thus, in the case of the DAMN polymer-SSP, three peaks have been established, one of which can be assigned to water evaporation. However, in the polymer sample produced by means of an MP, more peaks are clearly found. On the other hand, in the last decomposition stage, the two DTG peaks are located at temperatures above 650 °C, but the maximum decomposition rate for the sample obtained in the solid-state appears at lower temperatures in comparison with that synthesized from the melt, as can be clearly observed. The DSC thermograms in Fig. 10**b** do not exhibit significant differences, and mild endothermic peaks in stages I and III can be attributable to the wide variety of decomposition reactions occurring in this temperature range. Broadly, these decomposition processes generally involve deamination, decyanation, and other elimination reactions, as described based on the TG-MS data.

Based on the literature, thermal degradation of HCN polymers typically begins with an initial stage involving the release of trapped volatile materials such as water (m/z = 18) [55–58], and also with the decomposition of low molecular weight oligomeric species of poor thermal stability. Fig. 10**c** and **d** show the ion current for the main ions by mass (m/z) that were detected with MS as a function of the

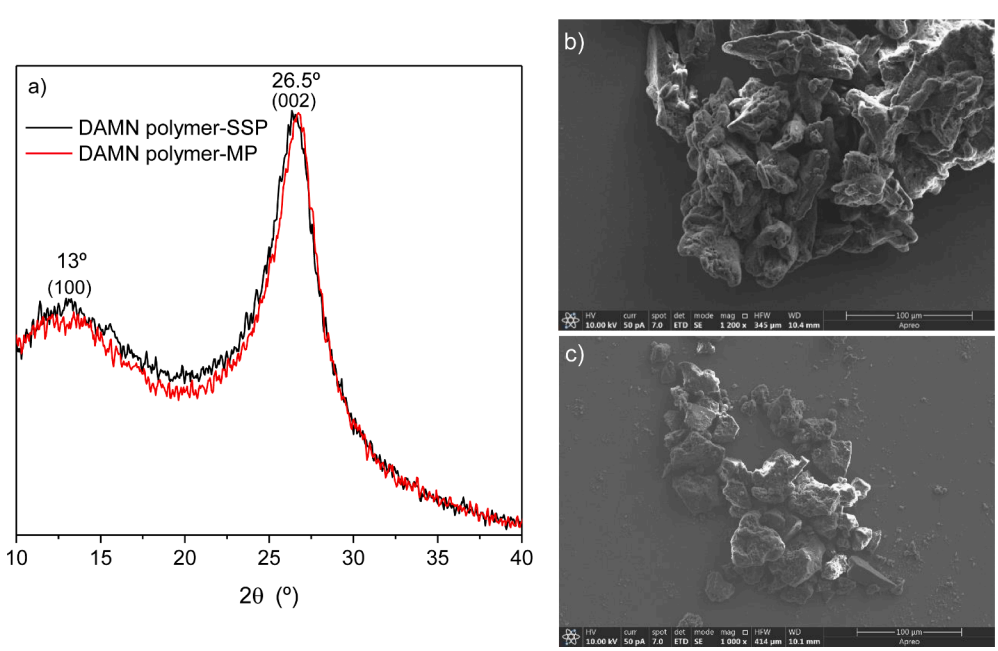


Fig. 9. a) XRD patterns; b) and c) SEM images of representative samples of DAMN polymer-SSP and DAMN polymer-MP, respectively.

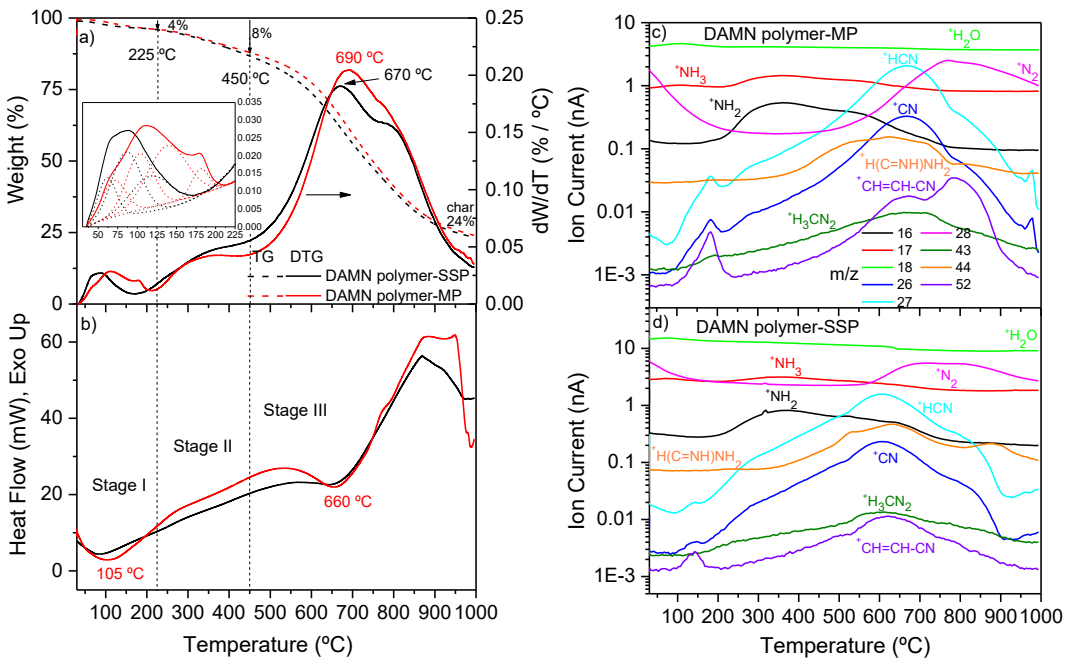


Fig. 10. a) TG and DTG curves; b) DSC curves; c) and d) ion intensity curves for the DAMN polymers MP and SSP, heating at 10 °C min^{-1} and argon atmosphere. Note that from the results obtained, it can be seen that the mixture of evolving gases is the same for both samples and that these mixtures are less complex than those from HCN-derived polymers obtained using aqueous conditions [55–58]. The lack of water during the polymerization processes ruled out the hydrolysis and oxidation reactions, simplifying the heterogeneity characteristic of the HCN-derived macrostructures.

temperature for both samples under study. The second major signal corresponding to $m/z = 28$ can be attributed to $^{+}N_2$. Note that the loss of N_2 has also been supported by thermal decomposition studies of carbon nitride materials at high pressures and temperatures [59], and because the loss of molecular nitrogen occurs mainly at a high temperature, at stage III above 500 °C, where thermal rearrangements can occur. The fragments with $m/z = 44$ can be recognized as formamidine $^{+}H(C = NH)NH_2$ and the corresponding deprotonated derivative $^{+}H_3CN_2$ for $m/z = 43$, taking into account the profile of both curves. Finally, the fragmented ion observed at $m/z = 52$ is proposed to eliminate $^{+}CH = CH$-CN, supported by the shape of the corresponding curve which is similar to those of the ions with $m/z = 26$ (^{+}CN) and 27 (^{+}HCN). In addition, it is significant that the three ion intensity variations can be observed over the three decomposition stages, showing a well-defined peak at higher temperatures. Therefore, these fragments must be from a fundamental part of the structural skeleton of both polymers.

Despite the minor thermal differences between the two polymers, MP synthesis seems to lead to a DAMN polymer that is more thermally stable. Generally, we have related this thermostability for HCN-derived polymers to a greater conjugation of the system and/or with a higher content of oxygenated functional groups [17,55]. Since the presence of oxygen is discarded in the present cases, one can think that the higher thermal stability of the DAMN polymer-MP is due to a greater conjugation of the system, which could be related to a greater amount of N-heterocycles in its structure, in agreement with the discussion in the above section.

3.5. Mechanistic considerations and hypothetical structures

Bearing in mind a global vision encompassing all these results, several mechanistic considerations can be addressed in an attempt to postulate likely macrostructures for the DAMN polymers explaining their redox properties and structural characteristics. Taking into account that DAMN is a tetrasubstituted alkene, and that this type of monomer presents a very low homopolymerizability through double bonds according to the studies of Hall and Padias [60], it seems that DAMN polymerization must occur by means of nitrile or imine groups on its tautomeric form, at least partially taking into account the significant number of nitrile groups that must be present on the backbone of these DAMN polymers.

In the case of DAMN, previous works have demonstrated that during its thermally induced polymerization weight loss is detected, as noted in Section 3.1 [15,16]. In this context, the first approach was to verify that during the isothermal polymerization of DAMN the two main elimination reactions, deamination (the removal of the amine group or ammonia, NH_3) and de(hydro)cyanation (the loss of cyanide ion ^{-}CN or hydrocyanic acid), are present. Thus, TG-MS data offer in-depth analytical support for in situ chemical characterisation of this thermally induced polymerization. Fig. 11 shows the main gaseous species generated and detected during isothermal polymerization at 190 °C. Again, liberation of both NH_3 ($^{+}NH_2$ $m/z = 16$, $^{+}NH_3$ $m/z = 17$ and

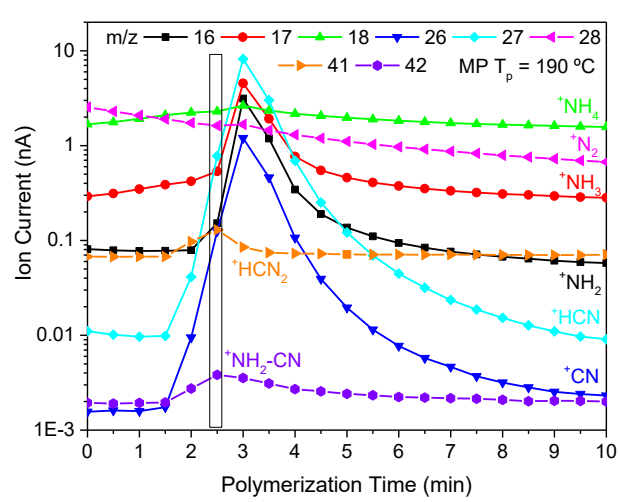


Fig. 11. Ionic current variation during the MP of DAMN at 190 °C under argon atmosphere.

$^{+}NH_4$ m/z = 18) and HCN (^{+}CN m/z = 26 and ^{+}HCN m/z = 27) is observed, along with other fragments reported in previous studies such as the ion with m/z = 42, $^{+}H_2CN_2$ cyanamide and its deprotonated derivative [15,16]. These two ions present their maximum slightly earlier than the rest of the signals.

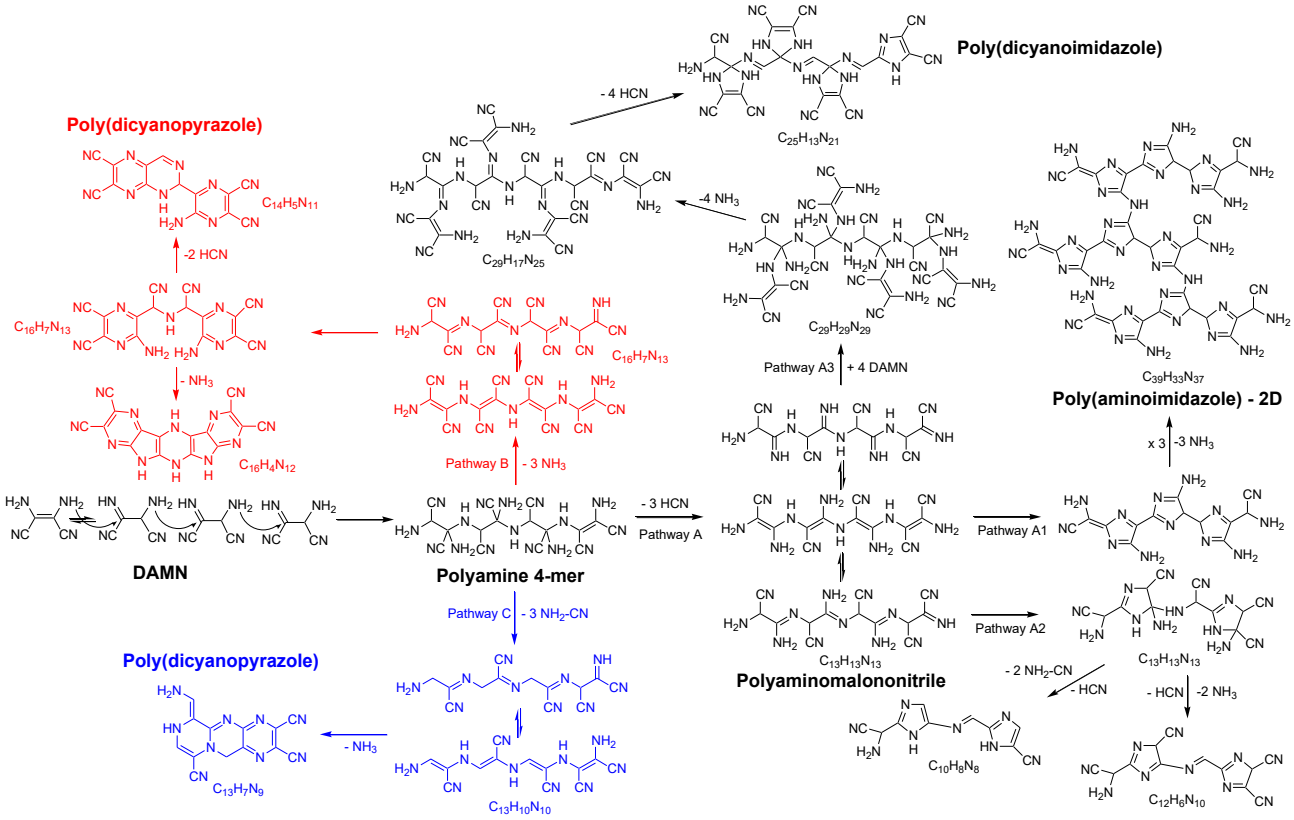
From these thermal results the following question arises: i) some elimination reactions on DAMN can give reactive precursors or intermediates of the polymerization, ii) in contrast, these small molecules are eliminated throughout the course of the polymerization reaction giving a highly conjugated polymeric system. Both cases are supported by the robustness and high efficiency of the bulk polymerization of DAMN, similar to a click reaction. However, the first of these hypotheses may be less probable considering the high yields found in both polymerization processes. For the second of these two possibilities, three different routes have been postulated and debated [3], but some of these hypothetical pathways are not considered here for several reasons. The first is that they only contemplate the loss of NH_3 during the polymerization mechanisms formulated. Second, both polymerization reactions involve the nitrile groups of DAMN and because of this the polyheterocyclics obtained practically lack cyano groups on their macrostructures.

In Scheme 1, a polyaddition is formulated by a reaction between the amine and imine groups of the DAMN tautomer. The approach is supported taking into account the high reactivity of the imine group, but the tautomeric equilibrium of the DAMN is predominantly expected in solution or in melt, and all the previous results suggest that bulk polymerization must go through a single mechanism regardless of whether the polymer reaction is in solid-state or melt. At this point, it is relevant to note that the π-conjugated system, present in the DAMN molecule, is responsible for changes in the reactivity of the amine group. In most aliphatic diamines, both amine groups react with electrophiles easily, such as carbonyl groups. However, experimental results have demonstrated that DAMN reacts with carbonyl compounds, giving only a mono Schiff base, and for this reason, the initial step formulated here contemplates only the reaction of an amine group [61].

Initially, a highly substituted polyamine, which must quite possibly evolve to a π-conjugated system, was obtained after the loss of HCN - Pathway A, NH_3 - Pathway B (red colour), or even NH_2 plus CN groups jointly - Pathway C (blue colour). Considering the dehydrocyanation reaction (Pathway A), this initial linear polymerization yields the so-called polyaminomalononitrile or protopeptide proposed by Matthews and Moser studying HCN chemistry in water [62]. In addition, other pathways have been previously suggested to yield this protopeptide based on the HCN trimer aminomalononitrile (AMN) by dehydrocyanation of DAMN, induced thermally. The AMN precursor may lead to the generation of dendrimeric macrostructures based on aminocyanoimidazoles with a globular shape [15]. However, these proposed structures are not in agreement with all the spectroscopic data discussed above and therefore could be discarded.

We think that these linear chains, with imine and enamine bonds decorating their skeleton, and their different tautomers with dynamic equilibria occurring easily, even in the solid-state, play a fundamental role in this polymerization. In addition, this macromolecular system must be sensitive to electronic and steric effects including π-conjugation and H-bonding, and its tautomers can evolve in different ways. For example, from this polyenamine the formation of a polyaminoimidazole has been previously formulated (Pathway A1) by reaction of pendant amine and nitrile groups, which was first proposed by Mamajanov and Herzfeld [26]. However, in a recent contribution we postulated that these heterocyclic chains or segments can evolve through continued elimination of NH_3, causing the formation of layers linked by polyimidazole units [15], as discussed below.

From the other tautomeric forms of the protopeptide, exhibiting a C = N moiety with electron-withdrawing properties resembling those of



Scheme 1. Hypothetical pathways for the formation of DAMN polymers based on their spectral features and the released gases detected by MS during the self-initiated thermal bulk polymerization of DAMN.

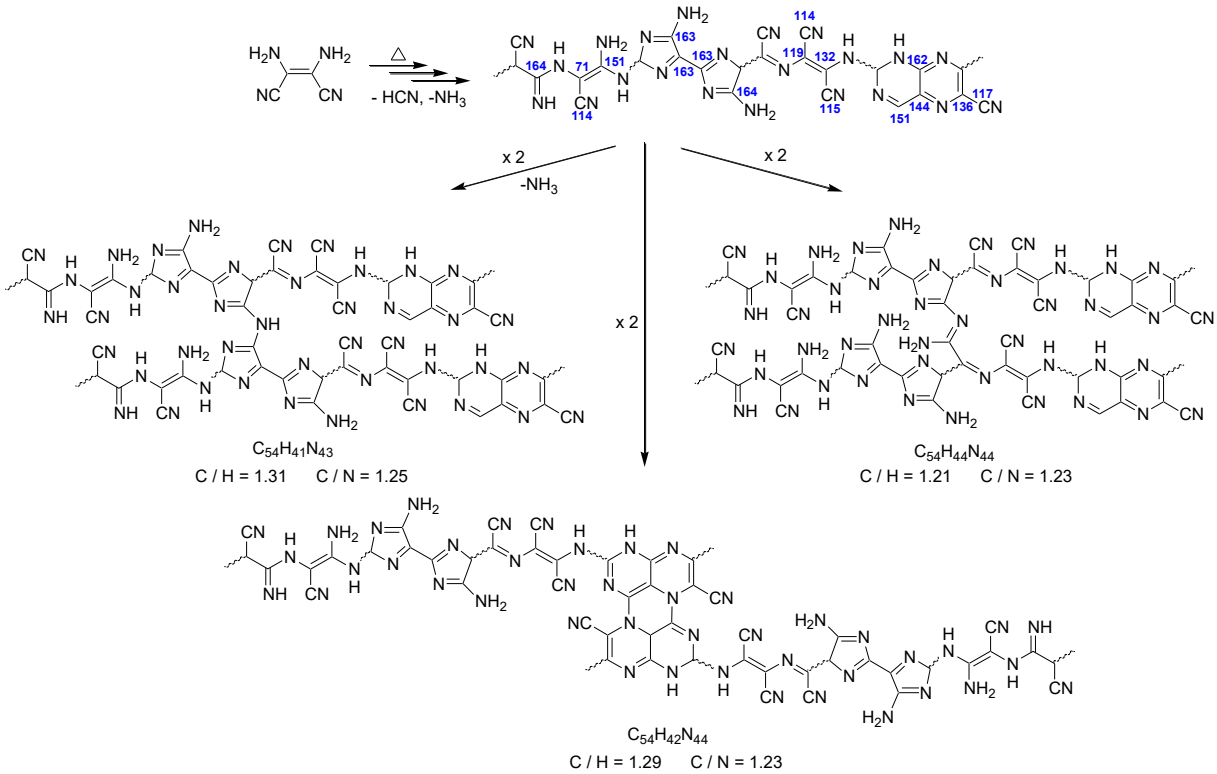
the carbonyl groups, it is possible to consider another route (Pathway A2). In this case, the intramolecular cyclization processes are between lateral amine groups and the imine groups on the backbone, with a higher reactivity of this C = N functionality in comparison with the cyano group. New polyimidazoles can be formulated after different elimination reactions. On the other hand, in Pathway A3 a nonlinear macromolecular system is formulated from this polyamidine due to its high reactivity through new additions of DAMN and successive ammonia elimination processes. This postulated conjugated macrostructure has not been considered until now, and can evolve in different ways, by intra- and intermolecular processes, and even produce aromatic CN heterocycles along its chains. In contrast, Pathway B allows the acquisition of polypyrazines by intramolecular cyclization between amine and nitrile groups, which can generate a ladder structure with fused rings. Finally, Pathway C, the polyenamine with nitrile pedant groups shown in this scheme, can also produce a polyheterocyclic system based on pyrazine rings similar to that formulated for Pathway B.

Taking into account Scheme 1, to achieve DAMN polymerization the system must be thermally activated. We have already tested this both in solution and in bulk under dynamic conditions [15,16,17]. This thermal activation would lead to a tautomeric equilibrium between the amine and imine forms of DAMN. Once the imine is formed, as it is the one that reacts, the system necessarily shifts the equilibrium towards its formation in such a way that an autocatalytic process is established where the generation of the initial polyamine would be the key to the subsequent steps of polymerization through the three proposed routes, A, B and C. The pathways in Scheme 1 seem to demonstrate the simultaneous loss of HCN, NH_3 and cyanamide; hence, from this global picture and with the great diversity of structures represented here, the most likely configuration of these DAMN polymers must be a rather complex macrostructure with different segments or blocks formed by heterocyclics, imidazole and pyrazole rings, and noncyclic subunits, supporting mainly imine but also enamine functional groups, as displayed in Scheme 2. The corresponding chemical shift δ (ppm) of the carbons in the simulated ^{13}C NMR spectra, calculated from ChemDraw Professional 15.0 software, can be observed together with the theoretical elemental analysis data. All these resonances can be observed in the spectra shown in Fig. 7c, and the calculated C/H and C/N values are close to the experimental data indicated in section 3.3, especially the macrostructure of the part on the right of this scheme. After a first polymerization step, where linear growth takes place with the corresponding intramolecular cyclization reactions, it is easy to think that this representative chain should progress to a 2-D structure by different pathways as indicated in Scheme 2, some of which have been previously described, by means of the loss of NH_3. This last step has also been proposed for the synthesis of polymeric g-C_4N_3 networks from different organic precursors such as dicyandiamide, melamine, urea and related heterocyclic arenes [53]. Two additional routes have been formulated, one by reaction between amine and cyano pendant groups and through the pyrazole rings, where a redox process could be involved; these 2-D macrostructures are supported by the XRD measurements.

Scheme 2 shows the likely mechanistic complexity of the thermal DAMN polymerization and can explain the redox properties of the DAMN polymers due to the extended π-conjugation of the macrostructures proposed. In addition, the better properties of the DAMN polymer-MP compared with the DAMN polymer-SSP as sensor can be related to a greater formation of N-heterocycles, probably increased by the higher temperature of reaction. However, due to the very similar structural and thermal characteristics found for both polymers the mechanistic pathways of formation in the solid-state and in the melt must follow the same main steps. Therefore, the difference in the reaction enthalpies between both polymerizations must possibly be related to a greater extension of the elimination and cyclization reactions as well as crosslinking in the MP.

4. Concluding remarks and outlooks

The thermal activation of DAMN, recognized as an important



Scheme 2. Representative structures for polymers based on DAMN obtained from its self-initiated thermal bulk polymerization. The blue numbers shown correspond to the chemical shift δ (ppm) of these carbons in the simulated ^{13}C NMR spectra using ChemDraw Professional 15.0 software.

prebiotic precursor, under plausible early Earth conditions, presumably leads to the formation of polymeric macrostructures constituted of linear polyimine blocks joined to N-heterocycle segments formed by imidazole and/or pyrazole rings, establishing multidimensional networks with good conductive properties. The electrochemical characterisation of electrodes modified with synthesized DAMN polymers demonstrates the presence of two monoelectronic redox systems with fast electron transfer corresponding to the same functional groups, although with different quantitative relationships. The results of the electrochemical and EIS studies indicate that the polymers obtained with MP have better electrocatalytic and conductive properties than those produced with SSP. This has allowed the use of the DAMN polymer-MP-modified electrode as an efficient and sensitive peroxide sensor, as an example, with good expectations for the preparation of biosensors with oxidase enzymes, but which might also be considered a protoenzyme due to its catalytic activity. In addition, the self-thermal polycondensation of DAMN both in the solid-state and in the melt, is an autocatalytic, powerful, and enormously efficient process similar to a click polymerization reaction and yields materials with good stability. The ease and robustness of these thermal polymerizations, inspired by low cost from prebiotic chemistry, encourage the development of a potential new generation of functional materials and the exploration of a pristine chemical space defined by a plausible primitive biochemistry.

CRediT authorship contribution statement

Marta Ruiz-Bermejo: Data curation, Formal analysis, Investigation, Visualization, Writing – review & editing. **Pilar García-Armada:** Data curation, Formal analysis, Investigation, Writing – original draft. **Eva Mateo-Martí:** Data curation, Formal analysis, Investigation, Writing – original draft. **José L. de la Fuente:** Conceptualization, Investigation, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors used the research facilities of the Centro de Astrobiología (CAB) and were supported by the Instituto Nacional de Técnica Aeroespacial "Esteban Terradas" (INTA), by the projects PID2019-104205 GB-C21 and PID2019-107442RB-C32 from the Spanish Ministerio de Ciencia e Innovación, and by the Spanish State Research Agency (AEI) project MDM-2017-0737 CAB (CSIC-INTA), Unidad de Excelencia María de Maeztu. The authors are grateful to Mª Teresa Fernández for generating the FTIR spectra and XRD measurements, to Pilar Valles from INTA for recording the SEM images, and students Cristina Pérez and Carlos Hortelano for their help in carrying out different measurements.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2021.110897.

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